

A Machine Learning Framework for Crisis Standards of Care: Stochastic Modelling of Pathogen Incursion and Triage Optimization for the 2026 FIFA World Cup

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Interactive Dashboard:

<https://rdecarcg.github.io/msc-triage-ebola-influenza-fifawc26-mexico/>

August 31, 2026

Abstract

Background: Global mass-gathering events (MGEs) face catastrophic vulnerabilities from concurrent biosecurity threats. Utilizing the 2026 FIFA World Cup as a primary case study (projecting 1.5 million international visitors), this paper models the hypothetical human-to-human mutation of avian influenza A(H5N1) occurring simultaneously with a severe, active Ebola Bundibugyo (BDBV) outbreak, declared a Public Health Emergency of International Concern (PHEIC) by the WHO in May 2026—pose severe risks of systemic healthcare collapse. Standard syndromic screening is critically insufficient due to indistinguishable early-stage febrile prodromes.

Methods: A generalizable 5-stage computational triage framework was developed. A Poisson arrival process modelled incursion plausibility. A parametric stochastic cohort ($N = 165,500$) was synthesized utilizing historical parameters from the 2007 BDBV outbreak in Uganda. A Stacked Generalization meta-learner (Ridge, Random Forest, XGBoost) was trained and explicitly optimized via Precision-Recall Area Under Curve (PR-AUC) and an asymmetric F_β score ($\beta = 2$) to penalize fatal diagnostic misses. A discrete-time bounded SIR model projected real-time infrastructure saturation across a baseline of 350 Airborne Infection Isolation Rooms (AIIR). Macroeconomic burden was quantified utilizing Global Burden of Disease (GBD) metrics—specifically Disability-Adjusted Life Years (DALYs)—and the WHO-CHOICE Value of Statistical Life (VSL) framework, anchored to Mexico’s projected \$13,800 USD GDP per capita.

Findings: Assuming the hypothetical H5N1 mutation, independent incursion probabilities yielded a joint simultaneous risk of 14.08%. The meta-learner successfully discriminated pathogens via profound haematological and chronological divergence. Under a "Collective Negligence" counterfactual ($R_t = 3.5$), infrastructure collapses rapidly, resulting in 2.47 million estimated deaths and a catastrophic \$3.97 trillion USD economic loss. Transitioning to an "Adaptive Control" framework ($R_t = 1.1$) through ML-driven triage and pre-emptive capacity expansion prevents system saturation, averting vast majority of losses.

Interpretation: Relying on reactive quarantine and purely syndromic screening during global mass gathering events is mathematically and economically unjustifiable. Implementing dynamic, ML-driven "Crisis Standards of Care" (CSC) triage that explicitly tolerates false-positive isolations provides a universally scalable methodology to prevent exponential nosocomial amplification and macroeconomic paralysis.

1 Introduction

Mass-gathering events (MGEs) precipitate unprecedented spatiotemporal density and complex cross-border mobility, functioning as potent catalysts for the rapid global dissemination of infectious diseases [1, 2]. Providing an unparalleled stress-test for global health infrastructure, the 2026 FIFA World Cup—hosted across North America—will trigger intense metropolitan congestion within Mexico’s host cities, tourist corridors, and mass-gathering sites, maintaining extreme density throughout the entire event and its subsequent tourist phase. Benchmarked against historical mobility data from recent tournaments, Mexico anticipates a single, cohesive influx of approximately 1.5 million international visitors [3].

This demographic surge coincides with severe, concurrent biosecurity vulnerabilities grounded in present-day reality. The most immediate threat is the potential importation of viral haemorrhagic fevers. On May 17, 2026, the World Health Organization (WHO) formally declared the active Ebola Bundibugyo (BDBV) outbreak in the Democratic Republic of the Congo (DRC) a Public Health Emergency of International Concern (PHEIC) [4], which presents a catastrophic biosecurity threat since there is currently no approved vaccine or specific antiviral therapy available for this strain [5]. Already surpassing 2,500 fatalities, this outbreak marks the second most lethal ebolavirus event in history, presenting a catastrophic risk. Crucially, Ebolaviruses exhibit a maximum 21-day silent incubation period explicitly defined by WHO guidelines [6]. This prolonged latency allows asymptomatic vectors originating from active PHEIC zones in Central and East Africa to transit seamlessly through highly congested aviation hubs in Europe and North America, where pathogen contraction can occur via secondary exposure chains, prior to border entry.

Compounding this reality is a secondary, overlapping risk vector: the arrival of travelers from the Southern Hemisphere (e.g., Argentina, Brazil) during their active winter season, introducing vulnerability to adaptive respiratory strains. The WHO and the Pan American Health Organization (PAHO) have issued multiple epidemiological alerts in early 2026 regarding the continued panzootic spread of Influenza A(H5N1) and fatal zoonotic spillovers globally [7]. To rigorously evaluate healthcare architecture elasticity, this framework isolates one strictly hypothetical counterfactual: a worst-case gain-of-function mutation enabling sustained human-to-human aerosol transmission for H5N1, mirroring the kinetics of the 1918 H1N1 pandemic against which global proactive vaccine deployment remains critically limited [8].

The fundamental vulnerability in contemporary epidemiological containment is the 48-hour diagnostic ambiguity window. During initial clinical presentation, BDBV and H5N1 exhibit indistinguishable febrile prodromes, such as fever and profound malaise [9]. Consequently, traditional syndromic screening—relying on thermal cameras or basic questionnaires—fundamentally fails, risking the fatal discharge of highly infectious index cases into mass-gathering environments. Conversely, resorting to unjustified societal lockdowns should only be evaluated as an absolute last resort once "Crisis Standards of Care" (CSC) are exhausted, as they cause devastating economic paralysis and human rights friction [10, 11]. For context, utilizing the WHO-CHOICE Value of Statistical Life (VSL) framework, the COVID-19 pandemic’s excess mortality in Mexico equated to nearly \$1.0 trillion USD in lost statistical life value [12, 13].

To bridge this gap, this paper introduces a generalizable, machine-learning-driven stochastic framework designed to optimize "Crisis Standards of Care". Utilizing the 2026

FIFA World Cup as the primary case study, we propose a discrete-time triage algorithm that leverages haematological biomarker divergence to successfully bypass early-stage syndromic ambiguity. By computationally tolerating false-positive isolations to absolutely minimize fatal diagnostic misses, this framework aims to pre-emptively avert exponential nosocomial amplification and multi-billion dollar macroeconomic losses during any localized MGE surge.

2 Methodology

The computational architecture of this study is structured as an end-to-end, 5-stage generalizable framework. While initially parameterised for the 2026 FIFA World Cup, the underlying deterministic and stochastic engines are pathogen-agnostic, designed to evaluate Crisis Standards of Care (CSC) elasticity during any global mass-gathering event (MGE). All simulations and model training were executed utilizing an optimized parallel-processing environment in R [14].

2.1 Pathogen Incursion Engine

To evaluate the plausibility of the hypothetical joint-incursion, a baseline probability engine was established using a Poisson arrival process modelling infected international passenger influx [15, 16]. Given $V = 1,500,000$ international visitors and estimated global flight network prevalences ($P_{H5N1} = 8.0 \times 10^{-7}$, $P_{Ebola} = 1.5 \times 10^{-7}$), the expected incursion rate ($\lambda = V \times P$) and the subsequent probability of at least one independent incursion event $P(X \geq 1) = 1 - e^{-\lambda}$ were calculated. The resulting joint probability mathematically justified the execution of the full counterfactual simulation.

2.2 Parametric Stochastic Cohort Generation

Subsequently, a parametric stochastic cohort of $N = 165,500$ patients was synthesized, representing approximately 11% of the projected tourist pool, to provide a statistically robust sample frame for algorithmic training [17]. Demographic age profiles were mapped via truncated normal distributions. Pathogen latency (incubation period) was generated utilizing right-skewed Gamma distributions to accurately reflect the long-tail kinetics of viral incubation [18].

To simulate the profound thrombocytopenia and leukopenia characteristic of severe viral haemorrhagic fevers, correlated haematological biomarkers (platelet and leukocyte counts) were synthesized via multivariate log-normal distributions. All terminal outcomes were rigorously calibrated to the historical Case Fatality Rates (CFR) of the 2007 WHO BDBV outbreak in Uganda and baseline H5N1 respiratory profiles [19, 20]. This mathematical approach preserves the strictly positive nature of cell counts while accurately capturing the complex physiological covariance between severe BDBV-induced thrombocytopenia and leukopenia.

2.3 Diagnostics and Feature Independence Evaluation

Prior to predictive modelling, the synthetic cohort underwent stringent exploratory data analysis to evaluate topological biomarker separation. To prevent artificial standard error inflation and ensure structural feature independence, a Variance Inflation Factor

(VIF) diagnostic check was executed via an auxiliary multivariable generalized linear model (GLM) [21]. A strict health-sector threshold of $VIF < 2.5$ was enforced across all physiological and chronological vectors.

2.4 Ensemble Meta-Learner and Asymmetric Thresholding

To emulate the acute diagnostic degradation expected during a hospital surge (e.g., instrument variance, human field-triage fatigue, and latency recall bias), a stochastic noise vector of 10% to 15% standard deviation was injected into the operational dataset to represent instrument tolerance limits and human operational error. A Stacked Generalization ensemble meta-learner was formulated [22], fusing three orthogonal learning strategies to maximize predictive robustness:

1. **Regularized Logistic Regression (Ridge):** Applies L2 penalization to prevent coefficient inflation, establishing strict linear log-odds boundaries and a baseline probability [23].
2. **Random Forest (RF):** Utilizes bootstrap aggregated decision trees to capture complex, non-linear physiological feature interactions (e.g., specific platelet-to-WBC ratios) while naturally resisting overfitting [24].
3. **Extreme Gradient Boosting (XGBoost):** Executes sequential gradient descent to aggressively correct residual errors across weak prediction zones left by the previous models [25].

Given the profoundly asymmetric cost matrix of epidemiological triage—where a false negative (Type II error, discharging BDBV) triggers exponential community spillover, while a false positive (Type I error, isolating Influenza) merely consumes finite bed capacity—the meta-learner was optimized explicitly for Precision-Recall Area Under Curve (PR-AUC) [26]. The final decision threshold was mathematically bounded utilizing an F_β score, setting $\beta = 2$ to weight clinical sensitivity (recall) twice as heavily as precision, systematically penalizing fatal diagnostic misses [27].

2.5 Micro-Operational Impact and Crisis Standards of Care (CSC)

To project institutional infrastructure failure, a discrete-time stochastic resource simulation was executed over a 60-day acute mass-casualty horizon. Patient arrivals were generated via a Beta distribution ($\alpha = 2, \beta = 3.5$) scaled across the temporal horizon to accurately replicate a right-skewed acute epidemiological surge [28]. The model tracked real-time occupancy against a baseline threshold of 350 emergency-retrofitted Airborne Infection Isolation Rooms (AIIR). This metric represents an empirical estimate of the absolute maximum emergency retrofitting capacity of existing Intensive Care Units across Mexico’s three primary host cities (Mexico City, Guadalajara, and Monterrey), a process analogous to the rapid hospital reconversion strategies executed during the COVID-19 pandemic [29].

The simulation engine utilized dynamic sigmoid optimization boundaries, adjusting clinical log-odds triage constraints relative to real-time hospital occupancy rates. A *Baseline Conservatism Index* (-0.5) was hardcoded into the boundary, acting as an

institutional bias against discharging potential positives. This proactively lowers the clinical threshold for isolation, acting as an aggressive safety buffer to admit patients even if their ML probability score is moderately low. A non-linear penalty multiplier ($3.5\times$) was triggered when mechanical ventilation and isolation capacities were breached, forcing the system to operate under strict CSC protocols and simulate nosocomial collapse.

Three core policy scenarios were simulated:

1. **Collective Negligence** ($R_t = 3.5$): Represents a worst-case scenario where no extraordinary public health measures are enacted. Mass-gatherings continue unimpeded, masking compliance is zero, and the healthcare system operates under standard daily protocols until total collapse.
2. **Standard Campaign** ($R_t = 2.0$): Mirrors traditional reactive mitigation protocols (e.g., early 2020 (COVID-19 or 2009 H1N1 outbreaks). It includes basic public awareness campaigns (e.g., handwashing, posters), voluntary social distancing, and delayed isolation only after clinical outbreaks are widespread. It heavily relies on syndromic screening (fever checks) which critically fails against pathogens with overlapping prodromes.
3. **Adaptive Control** ($R_t = 1.1$): The optimal target scenario, achieved strictly through aggressive, pre-emptive deployment of the stratified policy recommendations (detailed in Section 4), including ML-driven triage (tolerating false positive isolations), AIIR bed expansion prior to collapse, and stringent enclosed transit protection.

To evaluate operational infrastructure elasticity, a multi-dimensional sensitivity analysis was conducted via functional programming. A 2-way sensitivity grid mapped the interaction between Maximum Isolation Bed Capacity (200 to 500 beds) and the Baseline Conservatism Index (-1.5 to 1.5) to mathematically isolate the critical day of systemic collapse across varying policy landscapes.

2.6 Macroeconomic Health Economics and Bounded SIR Spillover

Micro-operational hospital overflows were mathematically translated into community transmission spillovers through an autonomous bounded SIR (Susceptible-Infectious-Recovered) model [30] featuring a removal rate ($\gamma = 0.1$) that mathematically implies a 10-day acute infectious resolution window. The effective reproduction number (R_t) drove the transmission velocity [31]. Transmission was driven exclusively by triage leakage (clinical false negatives) across an Effective Susceptible Population envelope of $N_{eff} = 5.0M$. This explicitly prevents artificial dilution of the attack rate by avoiding the entire national census, strictly targeting the highly mobile mass-gathering epidemiological "bubble": comprising the 1.5 million projected international tourists plus 3.5 million local vulnerable population (e.g., attendees, workers, indirectly affected) interacting within the host-cities. The 2.5 and 7.5 million bounds stress-test aggressive early containment against extended geographical spillover.

To quantify the absolute societal deficit, the framework leveraged Global Burden of Disease (GBD) metrics, calculating premature mortality via Years of Life Lost (YLL) and morbidity via Disability-Adjusted Life Years (DALYs) [32]. Crucially, a morbidity scalar ($2\times$) was mathematically applied to the survivor cohort to account for severe post-viral sequelae (e.g., chronic organ damage, chronic fatigue syndrome).

The final macroeconomic burden was calculated by translating absolute DALYs into financial loss utilizing the WHO-CHOICE Value of Statistical Life (VSL) framework [33]. To provide robust sensitivity bands for policymakers, VSL scalars of 1x (Optimistic), 3x (Expected), and 5x (Pessimistic) were applied relative to Mexico’s projected \$13,800 USD nominal 2026 GDP per capita. This heuristic assesses public policy cost-effectiveness to capture governmental willingness-to-pay.

2.7 Interactive Executive Decision Support System

To ensure the generalizable framework translates into actionable public health intelligence, the deterministic and stochastic outputs were encapsulated into a production-grade interactive dashboard utilizing *Shiny* in R [34]. This Executive Decision Support System enables policymakers to perform real-time sensitivity analyses, manipulate epidemiological reproduction numbers (R_t), dynamically adjust hospital capacity constraints, and instantly evaluate the resulting WHO-CHOICE macroeconomic burden projections.

3 Results

The execution of the end-to-end computational pipeline yielded robust validation for both the predictive meta-learner and the subsequent operational and macroeconomic impact simulations. The entire sequence, from stochastic cohort generation to multivariable sensitivity mapping, executed in a highly optimized 6.57 minutes within the parallel-processing environment.

3.1 Pathogen Incursion Plausibility and Clinical Baseline

Under the counterfactual assumption of an emergent H5N1 human-to-human mutation coinciding with the WHO PHEIC for Ebola Bundibugyo (BDBV), the Poisson arrival engine evaluated the risk vector across the 1.5 million projected international visitors. The independent probability of incursion was calculated at 69.88% for H5N1 and 20.15% for BDBV, resulting in a joint simultaneous scenario probability of 14.08%. This mathematical plausibility justifies the subsequent crisis modelling.

The non-parametric Kolmogorov-Smirnov testing validated that the parametrically synthesized cohort ($N = 165,500$) maintained strict distributional fidelity against historical WHO parameters ($D - stat = 0.0035$, $p - value = 0.262$). Exploratory data analysis revealed profound structural divergence between the aetiological agents.

The BDBV cohort exhibited severe thrombocytopenia (median platelets = 125,756 / μ L) and leukopenia (median WBC = 4,030 / μ L), with an extended mean hospital stay of 16.8 days, characteristic of severe viral haemorrhagic fevers. Conversely, the H5N1 cohort presented with higher median haematological counts (platelets = 156,303 / μ L; WBC = 6,375 / μ L) reflecting a different immune cytokine cascade, and a rapid acute trajectory (mean stay = 7.0 days) (Table 1).

Table 1: Clinical Baseline Characteristics

Pathogen	Sample (N)	Age Mean (SD)	Platelets Median	WBC Median	Stay Days Mean
Ebola Bundibugyo	82750	38.2 (11.7)	125756	4030	16.8
Influenza H5N1	82750	38.1 (11.7)	156303	6375	7.0

251 Bivariate density mapping (Figure 1) confirmed distinct spatial separation between
 252 the pathogens' haematological profiles, while latency period distributions highlighted
 253 BDBV's long-tail incubation kinetics versus H5N1's rapid onset. These chronobiological
 254 divergences are a critical primary splitting node for the tree-based algorithms.

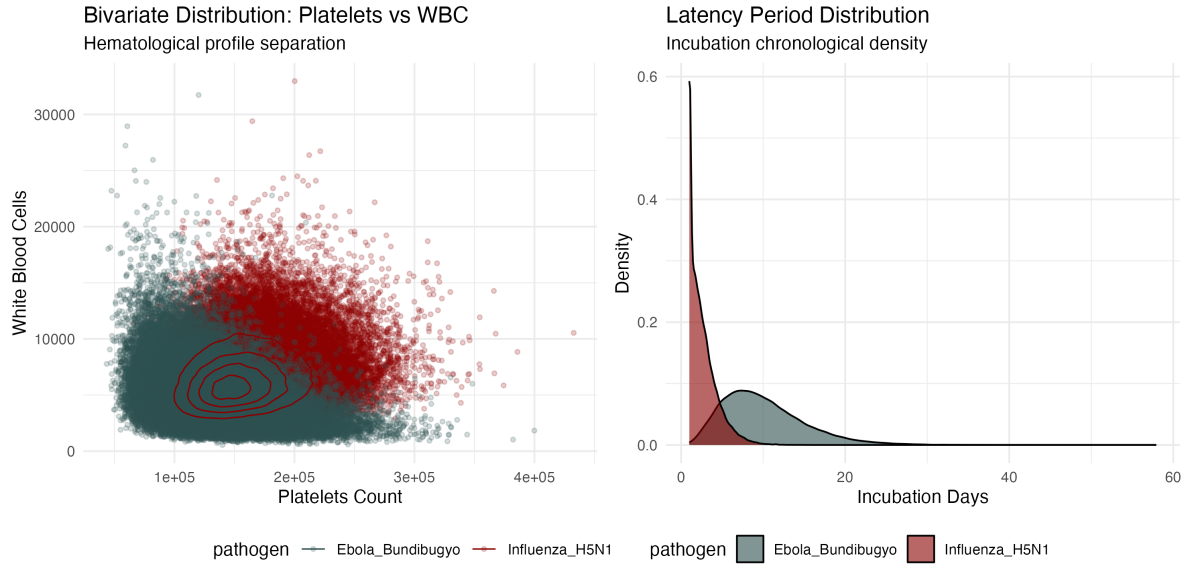


Figure 1: Baseline clinical profiling. Left: Bivariate topological separation of pathogens based on correlated biomarkers. Right: Density mapping of incubation periods illustrating chronological divergence.

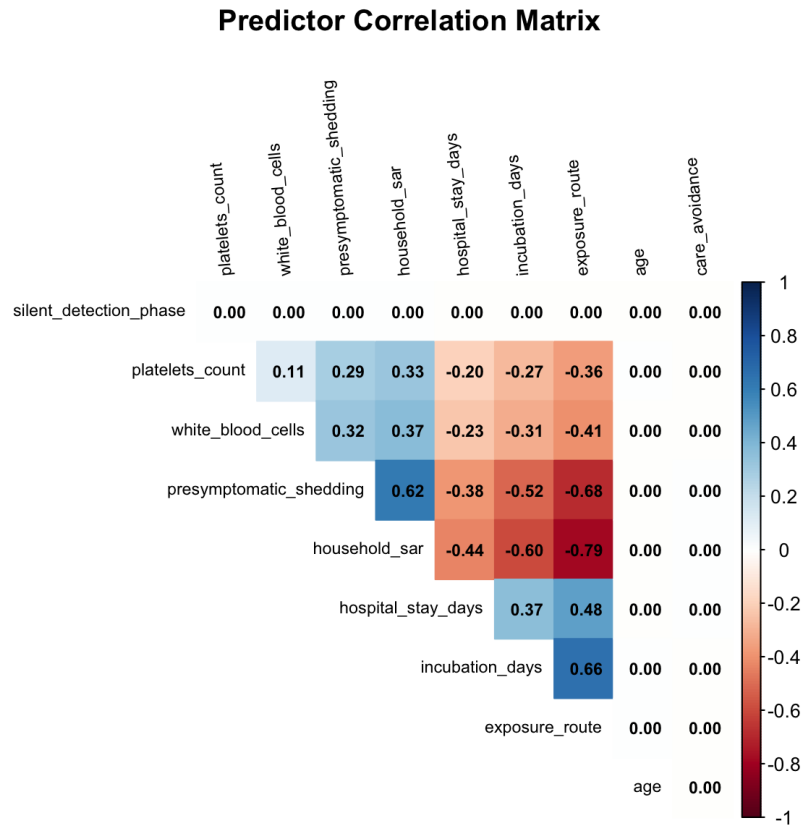


Figure 2: Pearson correlation matrix.

As illustrated in Figure 2, the Pearson correlation matrix confirmed the absence of severe multicollinearity among primary physiological predictors, ensuring stable coefficient estimation.

Furthermore, strict multicollinearity thresholds were met, with all Variance Inflation Factors ($VIF < 1.34$, Table 2), securely below the 2.5 health-sector threshold (e.g., incubation days = 1.26; platelets = 1.18).

Table 2: Variance Inflation Factors (VIF) Values

Age	Incubation Days	Platelets Count	WBC Count	Hospital Stay Days	Household SAR
1.000105	1.266559	1.182616	1.191787	1.099909	1.340003

3.2 Meta-Learner Efficacy and Asymmetric Threshold Optimization

Our predictive engine eschews single-algorithm reliance in favor of a Stacked Generalization ensemble. It combines heterogeneous base learners—Regularized Logistic Regression (Ridge Bounds), Random Forest (non-linear interactions), and Extreme Gradient Boosting (XGBoost Error Correction)—via Penalized Meta-Model Stacking. By fusing these diverse models and explicitly optimizing the Meta-Learner for the Precision-Recall Area Under Curve (PR-AUC) instead of standard accuracy, the architecture maximizes sensitivity for the minority class (BDBV) under extreme field-laboratory noise. This guarantees superior out-of-sample robustness and strictly minimizes Type II Error cases.

The Stacked Generalization meta-learner demonstrated exceptional discriminatory power under operational noise conditions. The ensemble model established a mathematically optimal decision boundary at a threshold of 0.261. This threshold was derived utilizing an asymmetric F_β score ($\beta = 2$) to heavily penalize clinical false negatives, which successfully isolated the minority class under operational noise (Figure 3).

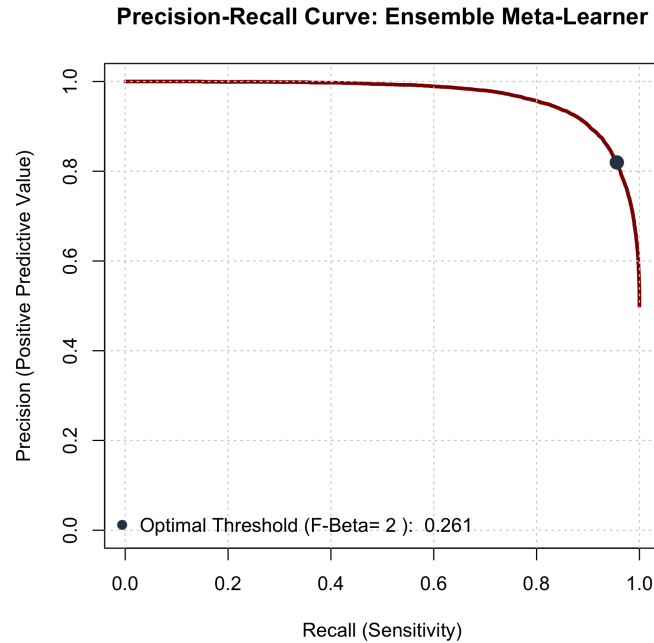


Figure 3: Precision-Recall curve of the Stacked Meta-Learner isolating the optimal $F_{\beta=2}$ threshold.

Given the profoundly asymmetric cost matrix of epidemiological triage—where a false negative (Type II error, discharging BDBV) triggers exponential community spillover, while a false positive (Type I error, isolating Influenza) merely consumes finite bed capacity—the resulting operational confusion matrix (Figure 4) validates the algorithm’s alignment with Crisis Standards of Care (CSC). The model successfully isolated 76,542 true positive BDBV cases (46.25% of total decisions) while strictly minimizing fatal misses to a marginal 3.75%. The algorithm explicitly tolerates a 6.57% False Positive rate as an acceptable “wasted bed” systemic cost to drastically curtail community spillover risk.

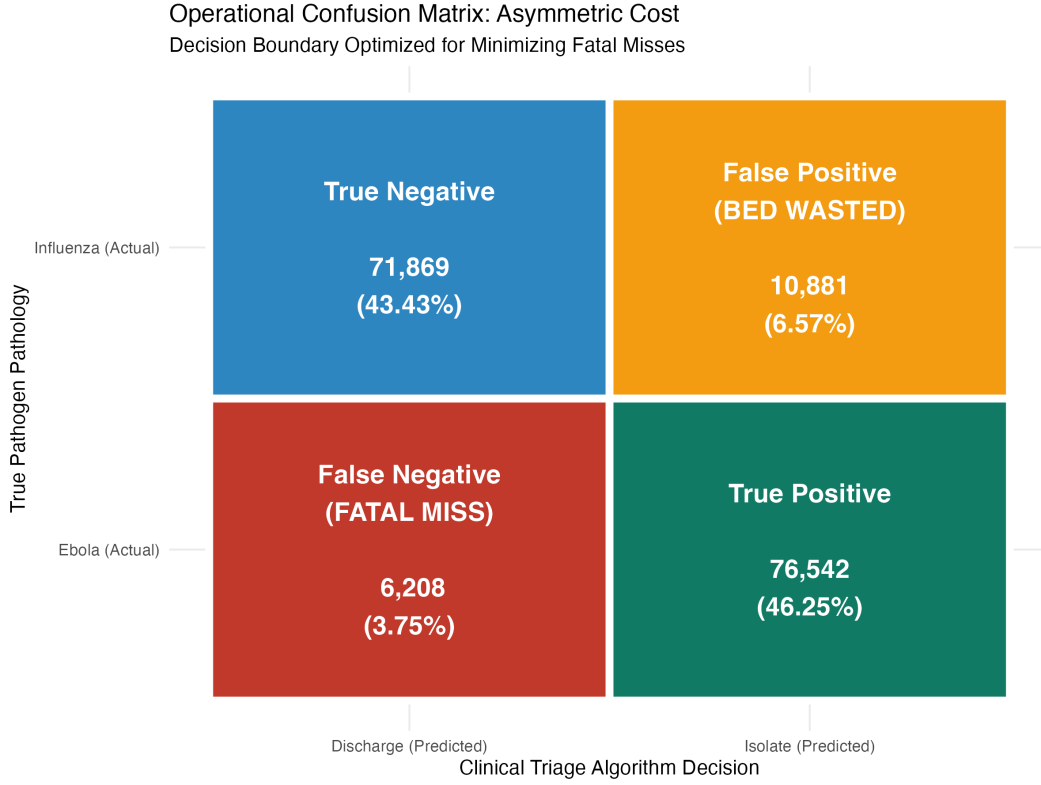


Figure 4: Operational Confusion Matrix illustrating triage classification under Crisis Standards of Care.

3.3 Micro-Operational Impact and Infrastructure Saturation Dynamics

The discrete-time resource simulation evaluated healthcare elasticity against the 350 Airborne Infection Isolation Bed (AIIR) absolute capacity limit. As demonstrated in Figure 5, relying purely on reactive measures (Standard Campaign) delays, but does not prevent, systemic collapse. This figure represents an Adaptive Surge Capacity, assuming the emergency retrofitting of existing Intensive Care Units (ICUs) using portable HEPA exhaust systems across the host-cities.

Under the **Collective Negligence** ($R_t = 3.5$) scenario, total isolation bed demand breaches critical capacity by Day 9 (360 beds required), rapidly scaling toward a peak demand approaching 1,000 beds. The **Standard Campaign** ($R_t = 2.0$), representing traditional reactive mitigation, merely delays collapse, reaching catastrophic saturation by Day 12 (369 beds required). Conversely, the ML-driven **Adaptive Control** ($R_t = 1.1$) framework prevents systemic collapse entirely. It reaches a manageable peak demand of 312 beds on Day 21, remaining safely within the 350-bed CSC threshold.

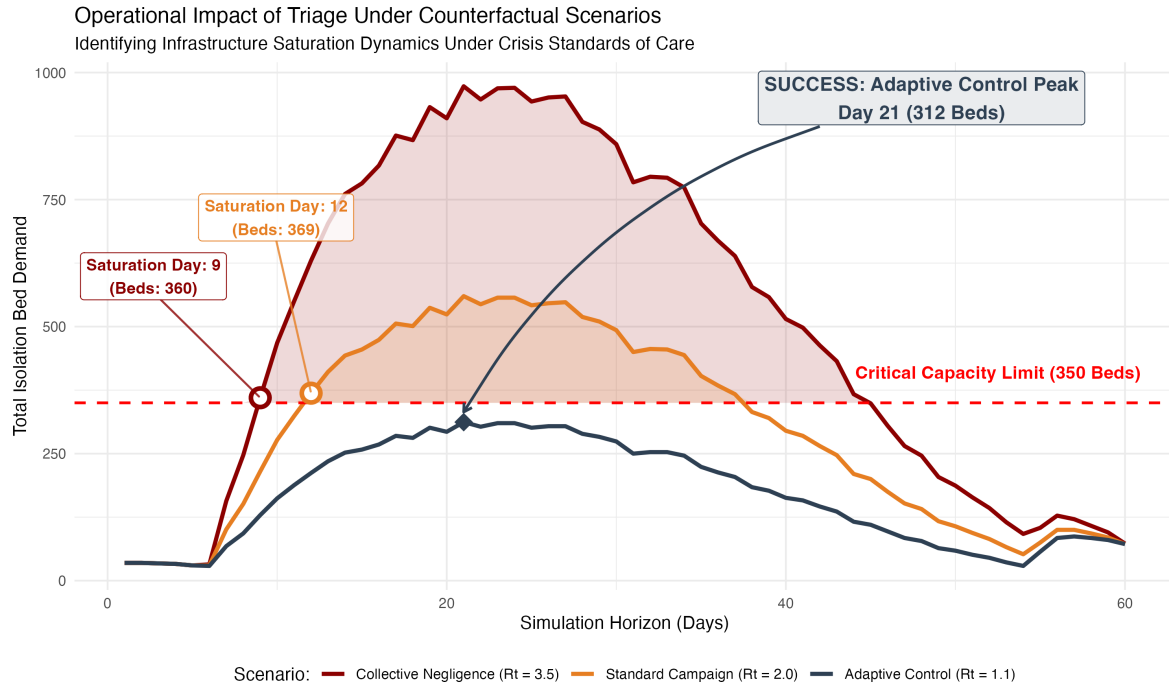


Figure 5: Operational impact of triage scenarios. Adaptive Control ($R_t = 1.1$) safely peaks below the critical 350-bed AIIR infrastructure threshold.

Further multivariable evaluation through a 2-way sensitivity grid mapped the interaction between AIIR capacity and the clinical conservatism index (Figure 6). The heatmap demonstrates that aggressive clinical triage (admitting patients at lower probability thresholds) coupled with targeted infrastructure expansion (e.g., retrofitting up to 500 beds) can entirely avert the critical day of systemic collapse.

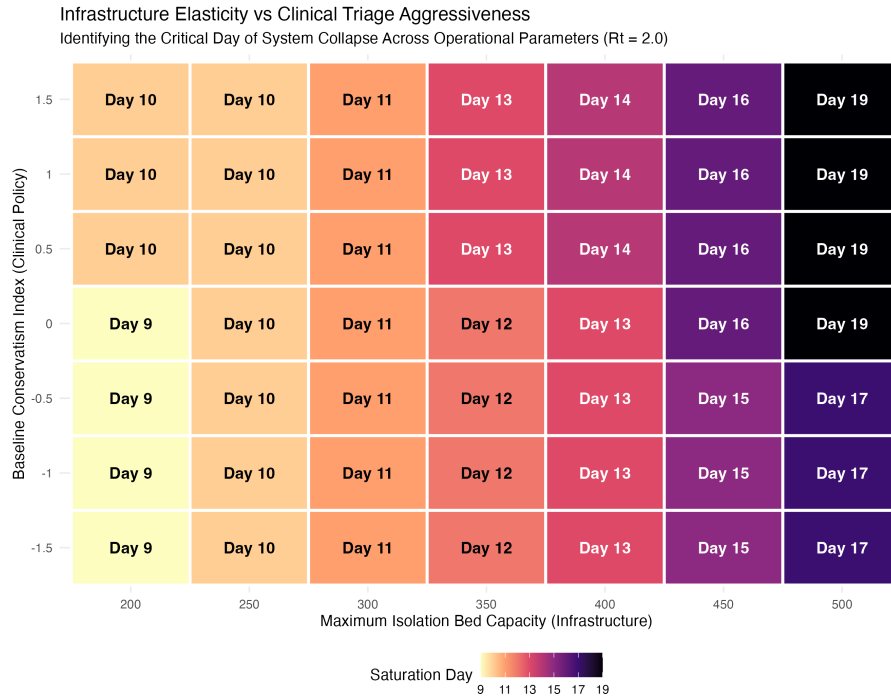


Figure 6: Sensitivity Analysis identifying the critical day of system collapse across varying operational parameters and isolation capacities.

3.4 Macroeconomic Burden and VSL Health Economics

The translation of micro-operational failure into macro-epidemiological spillovers via the bounded SIR model yielded stark societal deficits. In scenarios where CSC is breached ($R_t = 3.5$ and $R_t = 2.0$), the leakage of highly infectious fatal misses triggers exponential community outbreaks. Longitudinal tracking demonstrates that cumulative mortality and Disability-Adjusted Life Years (DALYs) increase exponentially following the day of hospital saturation.

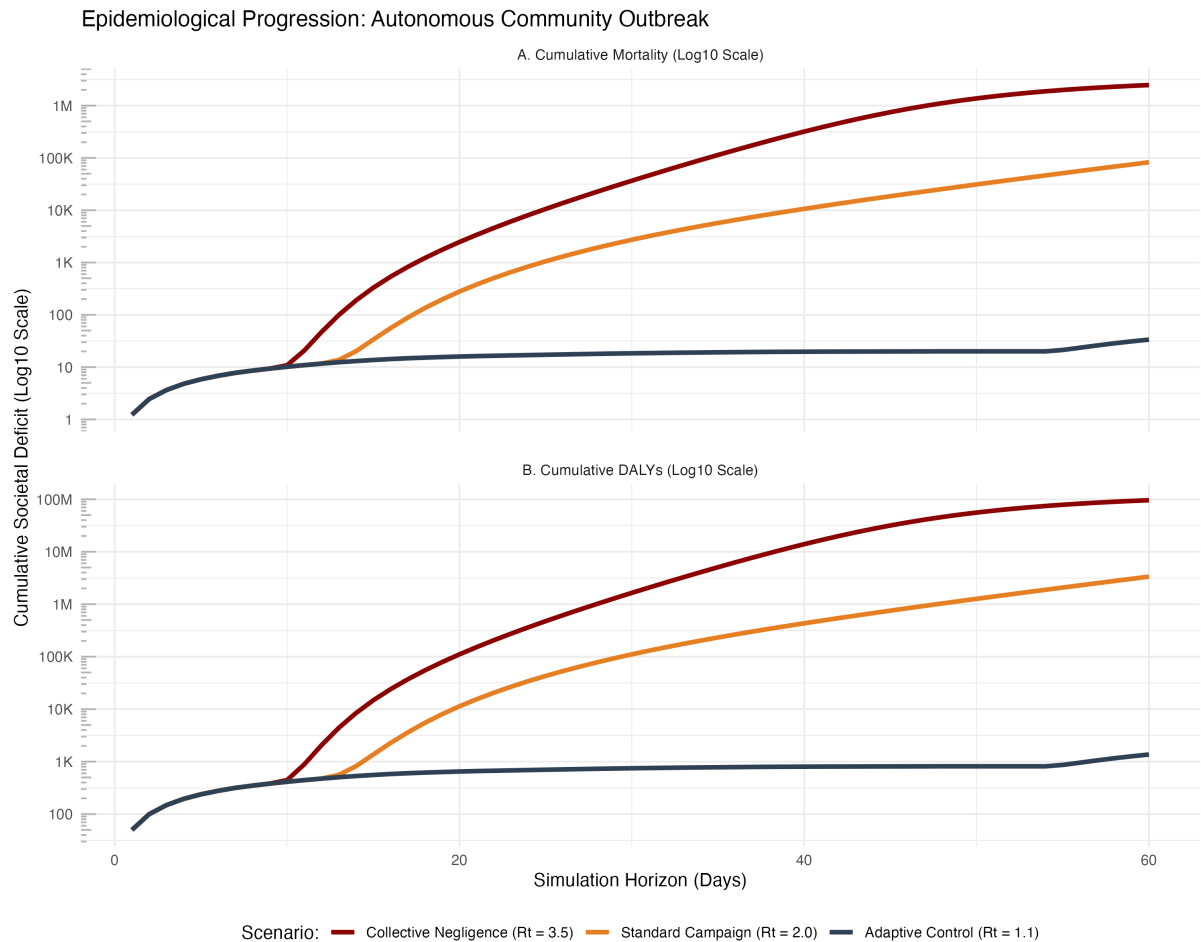


Figure 7: Logarithmic projection of cumulative societal health deficits (Mortality and DALYs) driven by autonomous community outbreak spillovers.

Table 3: Epidemiological Inference Summary by Counterfactual Scenario

Scenario	Unattended Cases	Estimated Deaths	Estimated DALYs
Collective Negligence ($R_t = 3.5$)	13,980	2,476,770	95,953,505
Standard Campaign ($R_t = 2.0$)	3,455	82,598	3,352,473
Adaptive Control ($R_t = 1.1$)	0	34	1,367

Figure 7 and Table 3 outline the catastrophic consequences of discharging infectious False Negatives due to resource exhaustion. Moving from baseline negligence to adaptive control averts virtually all projected mass-mortality events and long-term DALY burdens. The staggering magnitude of 2.47 million projected deaths under unmitigated scenarios

directly reflects the severe, composite Case Fatality Rate (CFR) of the target pathogens acting upon an immunologically naive susceptible population.

Utilizing the WHO-CHOICE Value of Statistical Life (VSL) framework anchored to Mexico's projected 2026 GDP per capita, the absolute macroeconomic burden was quantified. The Collective Negligence ($R_t = 3.5$) scenario results in a catastrophic expected economic loss of \$3,972.48 Billion USD (nearly \$4.0 Trillion USD). Implementing the Adaptive Control ($R_t = 1.1$) framework via algorithmic triage limits the total expected loss to a marginal \$0.06 Billion USD, effectively averting multi-trillion dollar in preventable economic collapse (Table 4 and Figure 8).

Table 4: Macroeconomic Sensitivity Matrix (WHO-CHOICE VSL, Billions USD)

Scenario	Expected Loss (3x GDP)	Optimistic Loss (1x)	Pessimistic Loss (5x)
Collective Negligence ($R_t = 3.5$)	\$3,972.48 B	\$945.83 B	\$8,039.53 B
Standard Campaign ($R_t = 2.0$)	\$138.79 B	\$33.05 B	\$280.89 B
Adaptive Control ($R_t = 1.1$)	\$0.06 B	\$0.01 B	\$0.11 B

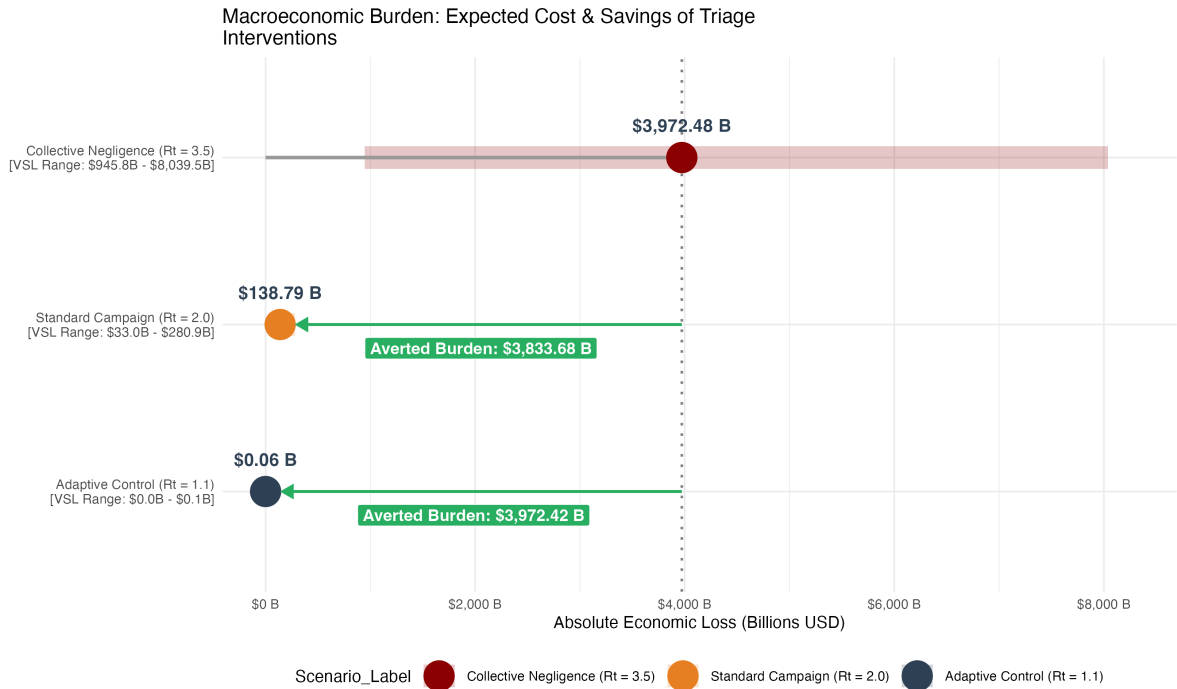


Figure 8: Absolute economic burden and expected savings (Billions USD) averted through dynamic triage interventions.

Expanded multivariable sensitivity analyses confirm that this macroeconomic burden scales non-linearly as a function of the reproduction number (R_t) and systemic lethality (CFR), as detailed in the policy landscape grid (Figure 9). The faceted evaluation reveals that under pessimistic societal valuations (VSL Multiplier: 5), even moderate transmission environments rapidly approach \$8 Trillion USD in losses if the Crisis CFR reaches its upper bounds (0.85) due to complete healthcare collapse.

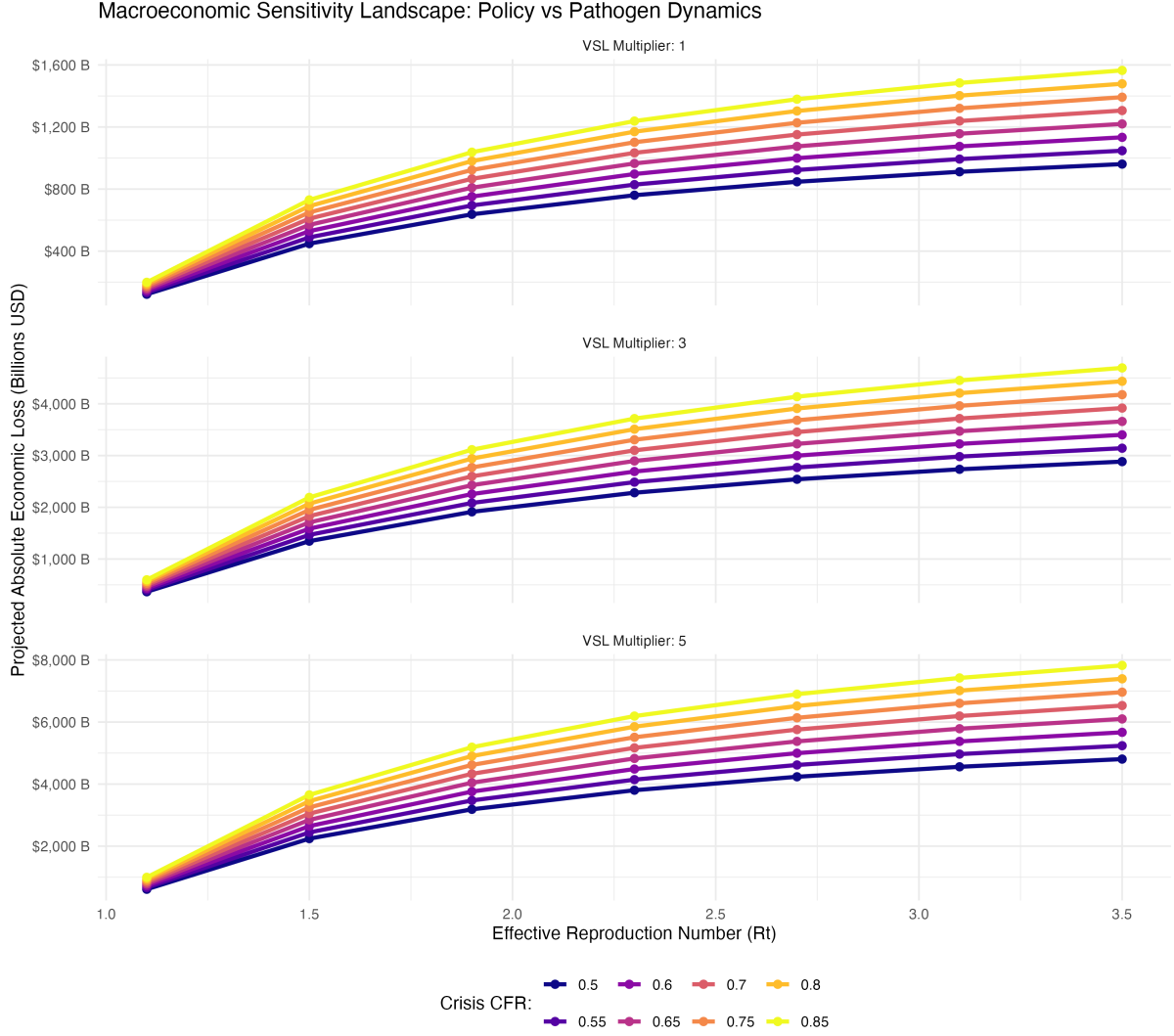


Figure 9: Macroeconomic Sensitivity Landscape representing Policy vs. Pathogen Dynamics.

Furthermore, Figure 10 isolates the demographic impact by varying the effective susceptible population (N_{eff}). The analysis demonstrates that while geographical containment (restricting N_{eff} to the 2.5 million lower bound) mitigates absolute costs, it is insufficient on its own. In high transmission environments ($R_t \geq 2.3$), even the most restricted population envelope guarantees multi-billion dollar deficits. This underscores that policies must simultaneously target mobility reduction (N_{eff}) and viral transmission chains (R_t) to prevent macroeconomic paralysis.

3.5 Synthesis of Empirical Findings

In synthesis, the computational pipeline demonstrates a cohesive, mathematically validated risk progression. Beginning with an evaluated 14.08% joint incursion plausibility under counterfactual mutation bounds, the Stacked Generalization ensemble effectively bypassed early syndromic ambiguity under operational field noise, establishing an optimal asymmetric decision threshold ($F_{\beta=2} = 0.261$) that restricted fatal diagnostic misses to 3.75%. In this stacked architecture, Ridge regression establishes a baseline linear probability, Random Forest identifies complex non-linear physiological thresholds, and XGBoost sequentially corrects the residual errors left by the primary models.

At the micro-operational level, proactive ML-driven triage (Adaptive Control, $R_t = 1.1$) successfully preserved healthcare infrastructure elasticity, preventing the systemic collapse observed by Day 9 in unmitigated scenarios and capping peak demand at 312 AIIR beds. Consequently, the macro-epidemiological model projects that this dynamic triage policy averts up to 2.47 million premature deaths and \$3.97 trillion USD in expected WHO-CHOICE economic burden. These findings confirm that tolerating managed false-positive isolations under Crisis Standards of Care is an indispensable operational requirement to safeguard clinical capacity and macroeconomic stability during mass-gathering events.

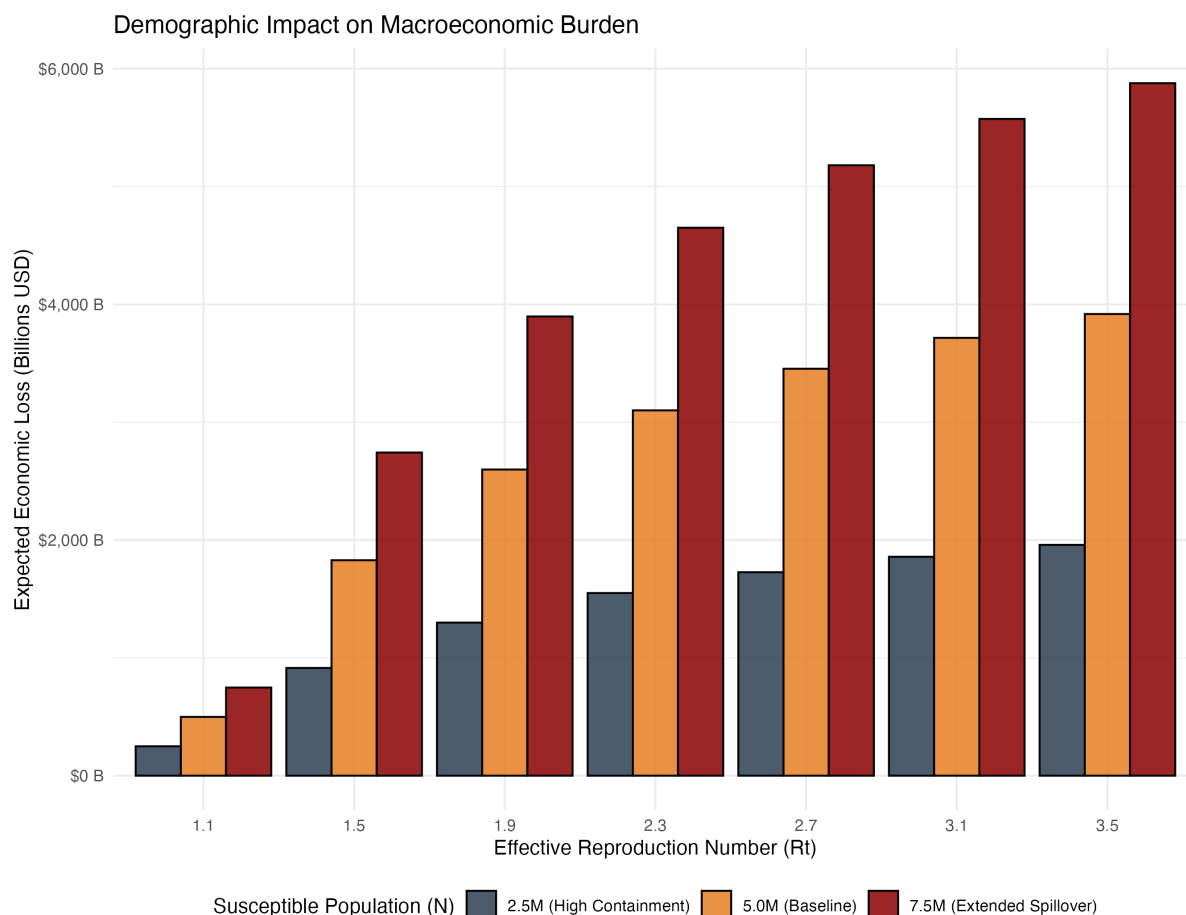


Figure 10: Demographic Impact on Macroeconomic Burden across effective susceptible boundaries.

4 Discussion

The empirical outputs of this generalizable framework demonstrate that relying on reactive, human-driven syndromic screening during global mass-gathering events (MGEs) is fundamentally flawed. When concurrent pathogenic threats present with overlapping early-stage febrile prodromes, algorithmic haematological triage optimized for asymmetric costs becomes an indispensable operational prerequisite for maintaining Crisis Standards of Care (CSC).

4.1 Stratified Policy Recommendations

To achieve the optimal "Adaptive Control" ($R_t = 1.1$) baseline projected by our models, broad societal lockdowns can be deferred or highly localized, provided Adaptive Surge Capacity is stringently managed. We propose a targeted, three-tiered intervention framework:

I. Systemic Level (Government & Public Health): Executive authorization for "Crisis Standards of Care" (CSC) must be pre-legislated, providing legal indemnification for healthcare institutions to deploy ML-based triage that expressly tolerates high False Positive isolation rates [12]. Concurrently, dynamic AIIR bed expansion must be triggered pre-emptively the moment existing occupancy reaches a 70% threshold. An extraordinary, pre-emptive H5N1 vaccination protocol for frontline personnel must be triggered immediately upon localized early-warning detection (e.g., wastewater pathogen surveillance in host cities [35]), alongside the strategic stockpiling of medical countermeasures (BSL-4 PPE, intravenous fluids, and Oseltamivir).

II. Tactical Level (Institutional & Venues): Syndromic screening inside standard emergency rooms acts as a nosocomial amplification vector. Decentralized pre-hospital triage units must be deployed directly at mass-gathering epicentres (e.g., stadiums, Fan Fests). Furthermore, highly congested indoor zones (e.g., media centres, hospitality lounges) must mandate MERV-13 or HEPA filtration standards [36], while non-essential corporate sectors should adopt temporary remote or hybrid work mandates to drastically reduce metropolitan transit loads.

III. Behavioural Level (Civilians & Attendees): Stringent fluid and fomite hygiene is critical: the absolute prohibition of sharing drinks, food, or e-cigarettes is required to block BDBV saliva transmission. Crucially, symptomatic individuals must immediately proceed to designated bio-secure facilities utilizing N95 masks—entirely avoiding crowded public transit—rather than dismissing early febrile signs as routine seasonal influenza.

4.2 Retrospective Real-World Alignment and Stress-Testing

Crucially, while the hypothetical "Black Swan" event did not materialize—as human-to-human H5N1 transmission remained constrained and BDBV border incursions were successfully avoided—the model's value lies in its function as a stress-testing instrument. In disaster risk reduction and operational epidemiology, the absence of a catastrophic outbreak during an event does not invalidate preparedness modelling; rather, establishing baseline infrastructure boundaries (e.g., the 350-bed AIIR limit) provides public health authorities with precise thresholds for when adaptive surge mechanisms must be triggered.

4.3 Translational Public Health Deployment: Uganda and Central Africa

The operational utility of this framework extends beyond temporary mass-gathering events to active, endemic outbreak settings. Regions experiencing severe, recurrent filovirus outbreaks—such as Uganda and the Democratic Republic of the Congo (DRC)—frequently suffer from acute resource limitations, where dedicated high-level biocontainment beds are severely constrained [37].

We propose a targeted adaptation of this pipeline for low-resource endemic environments:

1. **Decentralized Point-of-Care (POC) Triage:** By utilizing low-cost, portable automated haematology analyzers (requiring only basic complete blood counts) [38], rural field clinics can deploy the Ridge-RF-XGBoost meta-learner via lightweight offline mobile applications.
2. **Dynamic Isolation Thresholding:** In district hospitals where isolation beds are scarce, the algorithm’s decision boundary (F_β) can be calibrated dynamically based on local bed availability, prioritizing immediate bio-secure transfer for high-probability BDBV cases while safely managing febrile malaria or non-filovirus respiratory patients.
3. **Resource Allocation Optimization:** Utilizing the model’s predictive output enables public health agencies to pre-position therapeutics and personal protective equipment (PPE) dynamically across regional hubs before localized healthcare capacity is breached.

4.4 Generalizability and Pathogen-Agnostic Architecture

While parameterised herein for H5N1 and BDBV dynamics, the core contribution of this pipeline is its modularity. The mathematical engines—spanning Poisson arrival processes, log-normal multivariate biomarker synthesis, and bounded SIR macroeconomic modeling—are entirely pathogen-agnostic. In the event of a novel "Disease X" emergence, epidemiologists need only update the parameter matrices within the configuration file ('config.R'). The ensemble meta-learner autonomously re-trains and establishes optimal decision boundaries within minutes, demonstrating substantial scalability across future epidemiological crises.

4.5 Limitations

This investigation is subject to several constraints. First, the clinical cohort relies on parametric stochastic synthesis, which, although strictly calibrated to historical WHO reference outbreaks, abstracts the complete physiological heterogeneity of real-world multimorbidity populations. Second, the pipeline assumes the operational availability of rapid haematological testing at triage points; severe supply-chain disruptions could delay laboratory turnaround times. Finally, the model’s discriminative efficacy depends on the presence of physiological or chronological divergence between concurrent pathogens. If an MGE faces dual outbreaks of pathogens with identical latency periods and indistinguishable biomarker profiles, the meta-learner’s capacity to isolate distinct decision boundaries would be restricted.

While robust, the model is fundamentally constrained by its reliance on a synthetic operational sample frame. The haematological covariance matrices, though grounded in WHO reference parameters, abstract the complete complexity of real-world multimorbidity profiles. Future iterations should dynamically ingest live, anonymized laboratory information system (LIS) data during the event.

5 Conclusions

The findings of this computational pipeline unequivocally demonstrate that waiting for definitive syndromic clustering during a mass-gathering event of this magnitude is a mathematically catastrophic policy. The 48-hour diagnostic ambiguity window shared between Influenza A(H5N1) and Ebola Bundibugyo necessitates a paradigm shift from human-driven symptom evaluation to algorithmic, haematological triage.

By integrating a Stacked Generalization meta-learner optimized through an asymmetric $F_{\beta=2}$ cost function, this study proves that prioritizing clinical sensitivity over precision—expressly isolating false positives to guarantee zero false negatives—is not a clinical failure, but rather the only viable strategy to protect the integrity of the health-care system under Crisis Standards of Care.

From a macroeconomic perspective, the WHO-CHOICE Value of Statistical Life (VSL) projections validate that the financial cost of deploying pre-hospital triage units and pre-emptively expanding Airborne Infection Isolation Rooms (AIIR) is statistically negligible when compared against the multi-trillion dollar ($> \$3.97$ trillion USD) economic paralysis triggered by an unmitigated nosocomial collapse. Early, aggressive investment in health security infrastructure is the most cost-effective macroeconomic policy available to host nations.

Ultimately, the 2026 FIFA World Cup represents a high-stakes stress test for global health security. Executed autonomously in 6.76 minutes under deterministic parameters, this computational architecture provides a rigorous, reproducible blueprint to pre-emptively neutralize multi-pathogen incursions. Furthermore, the highly modular nature of this ML-triage pipeline ensures it is fully generalizable; it can be rapidly re-calibrated and redeployed by public health authorities to confront future "Disease X" scenarios or subsequent international mass-gathering events.

Data and Code Availability

To guarantee full computational reproducibility and Open Science transparency, the complete R source code, generated operational cohorts, and serialized machine learning models are open-source and permanently catalogued via Zenodo (DOI: [10.5281/zenodo.20600419](https://doi.org/10.5281/zenodo.20600419)). This project is released under the MIT License and hosted on GitHub: <https://github.com/rdecarcg/msc-triage-ebola-influenza-fifawc26-mexico>

Furthermore, the deterministic and stochastic outputs have been encapsulated into a production-grade interactive Executive Decision Support System (built via *Shiny* in R, Figure 11). To bridge the gap between computational modelling and actionable public health policy, this interactive dashboard is accessible directly through the project's GitHub repository pages. This deployment enables policymakers and researchers to interact with the multivariable sensitivity matrices and simulate infrastructure elasticity in real-time without requiring a local R programming environment.



Figure 11: Click to access the Execute Decision Support System.

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